6.8.1

b/d

Reflexive. A person has the same mother as herself or himself

Symmetric: if person x has the same mother as person y, then y has the same mother as x.

Transitive. IF x has the same mother as y and y has the same mother as z, then x has the same mother as z.

Each equivalence class consists of all the children of one mother.

(d) Reflexive: $x+x = 2x$. $2x$ is even so $x \mathrel{E} x$

Symmetric. IF $x+y$ is even, then $y+x$ is even

Transitive. Suppose that $x+y = 2k$ and $y+z = 2j$

there are two equivalence classes: set of all even integers and the set of all odd integers.

6.8.5

(b)

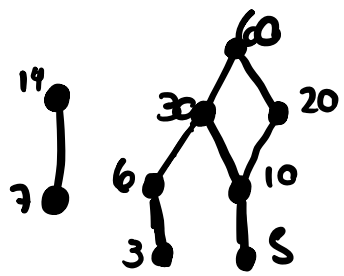
R is not an equivalence Relation

6.9.1 b

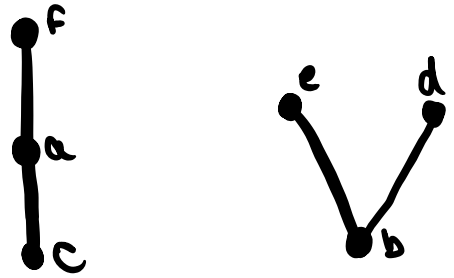
J, H, D, G

6.9.2 b/d

(b)



(d)



6.10.1 b/d

(b) Partial order BUT not a TOTAL order
Reflexive . x is SubsTring of x.

Anti- Symmetric. IF x is a substring of y and y is substring of z, Then x is also a substring of z.
Transitive. IF x is substring of y and y is a substring of z, then x is also a substring of z.

(d) Strict order but not total order

6.10.3 b
there is only one possible schedule. Quarter 1: Cs 10, Math 2A.
Quarter 2: Math 6A, Cs 11.
Quarter 3: Math 2B, Cs 12.
Quarter 4: Cs 51, Cs 160
Quarter 5: Cs 101, Cs 181

Chapter 7

7.1.1 b d f

(b)
Vertices 1, 2, 3, 4 and 6 are the neighbors of vertex 5

(d) Vertices 2 and 5 are both adjacent to vertex 3.

(f) Yes. $(1,2,5)$ and $(2,3,5)$ are both K_3 subgraphs of 6

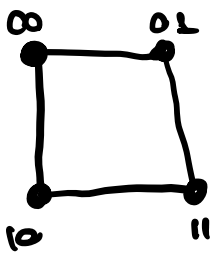
7.1.3 b

Yes. It is posible. For example $K_{3,3}$ is a 3-Regular graph with 6 vertices.

7.1.4 bd

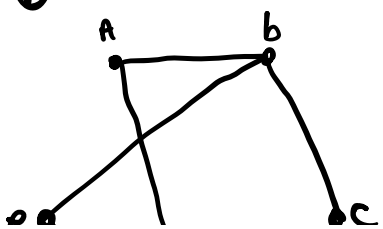
(b) There is an edge for every pair of vertices. So There are $\binom{5}{2} = 10$.

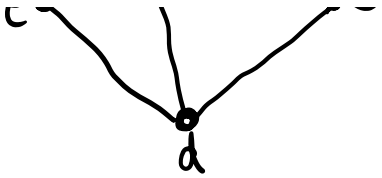
(d) $Q_2 = C_4$



7.2.1

(b)





7.2.2

(b)

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

7.2.3 (b)(d)

(b) Yes

(d) Yes

7.3.1 bdf

(b) Closed walk

(d) Closed walk, circuit

(E) Closed walk

7.3.2 bd

(b) 8. Vertex C can not participate in Cycle because there is only one edge incident to C.

(d) One possible solution ⟨B, E, B, A, B⟩

7.3.4 b

No, because the walk is a path, the walk cannot have Repeat vertices. Because the walk is a cycle, the first and last vertices must be the same.

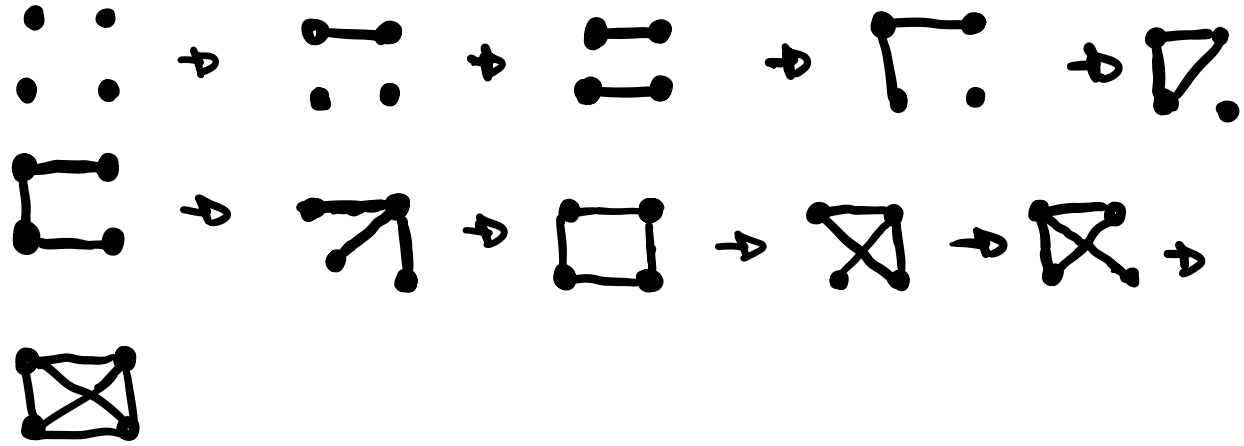
7.3.5 b

A path is a walk with no repeated vertices. the number of vertices that appear in a walk is at most n, the number of vertices in the graph.

7.5.2 b

the left graph has 5 vertices
the Right graph has 4 vertices

7.5.3 b



7.5.4 b

C_3 is a subgraph of the graph on the left.

7.6.1 b

5 components

$\{a\}, \{b\}, \{c\}, \{d\}, \{e\}$

7.6.2 b

Vertex Connectivity is 1

7.7.1 bd

2 vertices, b and F, have odd degree. Therefore the graph does not have an Euler circuit.

(d) Here is an Euler Circuit for the graph
Note that the solution is not unique.

7.7.2 b
the vertex w must be in C and G. The vertices that are

7.7.3 b/d
(b)
there are four vertices with odd degree: a, b, d and e. therefore the graph does not have an Euler Trail.

(d) the graph is not connected. therefore the graph does not have an Euler Trail.

7.7.4 b

C_n does not have an Euler Trail for any value of n.

7.8.1 b
Yes the path $\langle a, b \rangle$ is a Hamiltonian path

7.8.2 b

(b)

One possible solution: $\langle a, b, c, F, e, h, i, l, k, j, g, d, a \rangle$

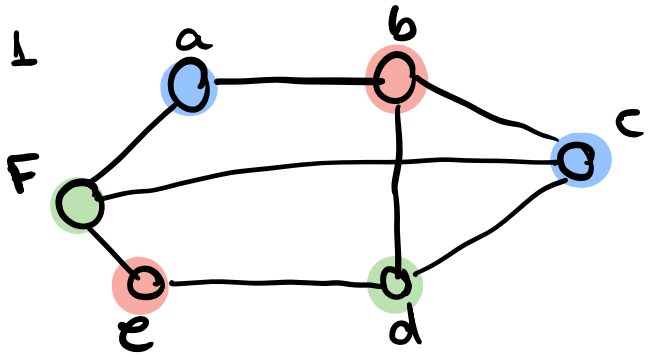
7.8.3 b

$\langle d, a, e, f, b, c \rangle$ is a Hamiltonian path in the graph there is an edge between every two consecutive vertices and every vertex appears exactly once.

7.9.1 b

5 regions.

7.10.1



7.6.2 bc

(b) The vertices of $K_{n,m}$ are divided in two sets.

(c)

Extra Credit

Therefore the shortest path from A to 2 is A-D-F, with distance = 11.